Implementation of the Burke 2012 Hydrogen Combustion Mechanism in Uintah's ICE Component

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1 Introduction

The following model is the implementation of the kinetic model for hydrogen oxidation proposed in *Comprehensive H2/O2 Kinetic Model for High-Pressure Combustion* by M. Burke et al. The following document describes the steps and equations used in order to simulate the combustion of hydrogen in air. The following assumptions and simplifications have been made:

1. ICE simulations are built on a single material. Individual species are not known by ICE only their representation as a passive scalar therefore phenomena like molecular diffusion due to composition change is not modelled.

2. Species ($H_2, O_2, N_2, H_2O, H, O, OH, HO_2, H_2O_2$) are tracked through the use of passive scalars. Passive scalars are passive with no affect on the physics of the flow. Each passive scale takes on a value $0 < \phi < 1$ which represents the mass fraction of each species. Only 8 passive scalars are tracked since the last species mass fraction can be computed from closure.

3. Thermodynamic data (h, g, c_p) is calculated using the NASA 7 polynomials. Model coefficients can be found in the mechanism file on the authors website: https://burke.me.columbia.edu/content/mechanis and-downloads

4. Finally this model only modifies the source terms of the governing PDE's, that being the energy source term and passive scalar source terms. The mixture specific heat as a function of temperature is also modified.

2 Reactions

The following reactions and related parameters are directly taken from the original mechanism. Some reactions have been omitted due to lack of species concentration in idealized air (Reactions with species Ar, He, CO, CO_2). Reaction numbering is based on the Cantera file and may have slight differences to the official publication.

Reactions with the terms $+M$ are third body reactions and are discussed in the entitled section. Reactions with the terms $(+M)$ are pressure dependent reactions and are discussed in the Troe Falloff Reactions section.

	Reaction	A	n	E_a	**Efficiencies** (Default 1.0)
(1)	$H + O_2 \leftrightarrows O + OH$	1.04e14	0.0	1.5286e4	
(2)	$O + H_2 \leftrightarrows H + OH$	3.818e12	0.0	7.948e3	
(3)	$O + H_2 \leftrightarrows H + OH$	8.792e14	0.0	1.917e4	
(4)	$H_2 + OH \leftrightarrows H_2O + H$	2.16e8	1.51	3.43e3	
(5)	$OH + OH \leftrightarrows O + H_2O$	3.34e4	2.42	-1.93e3	
(6)	$H_2 + M \leftrightarrows H + H + M$	4.577e19	-1.4	1.0438e5	$H_2 : 2.5;\ H_2O : 12$
(9)	$O + O + M \leftrightarrows O_2 + M$	6.165e15	-0.5	0	$H_2 : 2.5;\ H_2O : 12$
(12)	$O + H + M \leftrightarrows OH + M$	4.714e18	-1.0	0	$H_2 : 2.5;\ H_2O : 12$
(13)	$H_2O + M \leftrightarrows H + OH + M$	6.064e27	-3.322	1.2079e5	$H_2 : 3;\ O_2 : 1.5;\ N_2 : 2;\ H_2O : 0$
(14)	$H_2O + H_2O \leftrightarrows H + OH + H_2O$	1.006e26	-2.44	1.2018e5	
(15_L)	$H + O_2(+M) \leftrightarrows HO_2(+M)$	6.366e20	-1.72	5.248e2	$H_2 : 2;\ O_2 : 0.78;\ H_2O : 14$
(15_H)		4.65084e12	0.44	0	
(16)	$HO_2 + H \leftrightarrows H_2 + O_2$	2.75e6	2.09	-1.451e3	
(17)	$HO_2 + H \leftrightarrows OH + OH$	7.079e13	0.0	2.95e2	
(18)	$HO_2 + O \leftrightarrows O_2 + OH$	2.85e10	1.0	-7.239e2	
(19)	$HO_2 + OH \leftrightarrows H_2O + O_2$	2.89e13	0.0	-4.97e2	
(20)	$HO_2 + HO_2 \leftrightarrows H_2O_2 + O_2$	4.2e14	0.0	1.1982e4	
(21)	$HO_2 + HO_2 \leftrightarrows H_2O_2 + O_2$	1.3e11	0.0	-1.629e3	
(22_L)	$H_2O_2(+M) \leftrightarrows OH + OH(+M)$	2.49e24	-2.3	4.8749e4	$H_2 : 3.7;\ O_2 : 1.2;\ N_2 : 1.5;$
(22_H)			0.9	4.8749e4	$H_2O : 7.5;\ H_2O_2$
(23)	$H_2O_2 + H \leftrightarrows H_2O + OH$	2.41e13	0.0	3.97e3	
(24)	$H_2O_2 + H \leftrightarrows HO_2 + H_2$	4.82e13	0.0	7.95e3	
(25)	$H_2O_2 + O \leftrightarrows OH + HO_2$	9.55e6	2.0	3.97e3	
(26)	$H_2O_2 + OH \leftrightarrows HO_2 + H_2O$	1.74e12	0.0	3.18e2	
(27)	$H_2O_2 + OH \leftrightarrows HO_2 + H_2O$	7.59e13	0.0	7.27e3	

3 Reaction Rates

In order to calculate a reaction rate the following parameters need to be computed. A forward reaction rate k_f, an equilibrium constant k_c, and a reverse reaction rate k_r. A majority of the reactions (16) follow the common formula $R_1 + R_2 \leftrightarrows P_1 + P_2$ where R stands for reactant and P stands for product. Therefore to calculate the reaction rate (the rate at which the concentration of any species in the reaction is changing) for this type of reaction, the following formula is used:

$$\omega = k_f[R_1][R_2] - k_r[P_1][P_2]$$

Where [] indicate the concentration of that species. To calculate the forward reaction rate, modified Arrhenius law is used:

$$k_f = AT^n \exp(E_a/RT)$$

With the constants for each reaction defined in the table. For a balanced bimolecular reaction, an equilibrium constant is defined as:

$$k_c = \frac{k_f}{k_r} = k_p$$

Where

$$k_p = \exp(-\Delta G^o/RT)$$

Which is the exponential of the reactions nondimensional net Gibbs free energy. While all reactions are written as reversible, it is entropy contribution in Gibbs free energy that ultimately drives reactions from reactants to products in the direction of increasing thermodynamic favorability (i.e., toward equilibrium). With k_c and k_f, k_r can easily be calculated completing the parameters needed for calculating the reaction rate.

3.1 Duplicate Reactions

Some reactions are seen written twice with different Arrhenius parameters (i.e. 2, 3). These are treated as one kinetic reaction with one rate defining the species creation / destruction and energy release. The only modification when it comes to dealing with these reactions is the forward rates are calculated as the sum of two Arrhenius laws. In the case of reactions 2 and 3:

$$k_{f,23} = A_2 \exp(E_{a,2}) + A_3 \exp(E_{a,3})$$

All other terms are then calculated with the modified forward rate.

4 Third Body Reactions

Third body reactions capture the chemistry other species not present in the reaction might be contributing. This is accomplished through an additional term M and is defined as the dot product of every species concentration in the mixture and it's corresponding efficiency coefficient defined in the table for each reaction.

$$M = \sum_i \alpha_i [s_i]$$

With the third body term, the reaction rate can easily be calculated in a similar fashion as above. While third body reactions can be balanced (equal mols of products to reactants), the reactions present are not and split into the following categories.

4.1 2 Reactants, 1 Product

For these third body reactions, the reaction rate is calculated as follows:

$$\omega = k_f [R_1][R_2][M] - k_r [P_1][M]$$

However in order to keep consistent dimensions, the reverse reaction rate must be calculated from an equilibrium constant that has a correction term that in the previous section simplified to 1. In this case:

$$k_c = \frac{k_f}{k_r} = k_p \frac{RT}{P_{\text{ref}}}$$

4.2 1 Reactant, 2 Products

For these third body reactions, the reaction rate is calculated as follows:

$$\omega = k_f [R_1][M] - k_r [P_1][P_2][M]$$

With the equilibrium constant having a similar correction factor to keep units consistent.

$$k_c = \frac{k_f}{k_r} = k_p \frac{P_{\text{ref}}}{RT}$$

5 Troe Falloff Reactions (Pressure dependent)

Pressure plays a very important role in kinetics so two pressure dependent reactions are included in this kinetic description. In these pressure dependent reactions, an effective forward rate is calculated from an interpolation from a high and low pressure limit. The effective forward rate is then used as any other forward rate in the **Third Body Reactions** section. The procedure for finding $k_{f,\text{eff}}$ is as follows:

Calculate the low (k_0) and high (k_∞) pressure limits using modified Arrhenius law defined above. Also calculate the third body term (M) as defined above. Find a reduced pressure:

$$P_r = \frac{k_0[M]}{k_\infty}$$

And a Troe Center Factor:

$$F_c = (1-a)\exp(-T/T_3) + a\exp(-T/T_1)$$

Next calculate some constants:

$$\begin{aligned} C &= -0.4 - 0.67\log_{10}(F_c) \\ N &= 0.75 - 1.27\log_{10}(F_c) \\ d &= 0.14 \end{aligned}$$

Find the Troe Falloff factor (F) using:

$$\log_{10}(F) = \frac{\log_{10}(F_c)}{1 + \left(\frac{\log_{10}(P_r)+C}{N-d(\log_{10}(P_r)+C)}\right)^2}$$

Finally find the effective forward rate:

$$k_{f,\text{eff}} = k_\infty \frac{P_r}{1+P_r} F$$

6 Functions

6.1 enthalpy(dbl T, int R1, int P1, int* R2, int* P2)

Purpose: Calculate the net enthalpy of a reaction.
Inputs: Temperature (T), Reactant 1 (R1), Product 1 (P1). If given: address of Reactant 2 (R2), address of Product 2 (P2).
Steps:

1. `if (T > T_mid)`: select high-temperature coefficients, else low-temperature
2. Define lambda: `h = [](sp) { return a0[sp]*T + a1[sp]*T^2 + ... + a5[sp]; }`
3. `h_R1, h_R2, h_P1, h_P2` ← compute enthalpy for each species; set `= 0` if not provided
4. `return` $R_u(h_{R1} + h_{R2} - h_{P1} - h_{P2})$

6.2 gibbs(dbl T, int R1, int P1, int* R2, int* P2)

Purpose: Calculate the net negative gibbs free energy of a reaction.
Inputs: Temperature (T), Reactant 1 (R1), Product 1 (P1). If given: address of Reactant 2 (R2), address of Product 2 (P2).
Steps:

1. `if (T > T_mid)`: select high-temperature coefficients, else low-temperature
2. Define lambda: `g = [](sp) { return a0[sp]*(1-ln(T)) - a1[sp]*T - ... + a5[sp] / T; }`
3. `g_R1, g_R2, g_P1, g_P2` ← compute gibbs free energy for each species; set `= 0` if not provided
4. `return` $(g_{R1} + g_{R2} - g_{P1} - g_{P2})$

6.3 cpSpecificHeat(dbl T, int n)

Purpose: Calculate the constant pressure specific heat for n species.
Inputs: Temperature (T), Number of species (n).
Steps:

1. `if (T > T_mid)`: select high-temperature coefficients, else low-temperature
2. `for (i = 0; i < n; i++) cp[i] = a0[i] + a1[i]T + ... + a4T^4` ← compute specific heat for each species
3. `return` *cp*

6.4 reaction(dbl T, dbl RT, vec C, int recNum, int R1, int R2, int P1, int P2)

Purpose: Calculate the reaction rate for a standard bimolecular reaction $R_1 + R_2 \leftrightarrows P_1 + P_2$.
Inputs: Temperature (T), R_uT (RT), species concentrations (C), reaction number (recNum), reactant indices (R1, R2), product indices (P1, P2).
Steps:

1. Decrement `recNum` for 0-based array indexing
2. $k_f = A \cdot T^n \cdot \exp(-E_a/RT)$ — forward rate via modified Arrhenius
3. $k_p = \exp(\texttt{gibbs}(T, R_1, P_1, \&R_2, \&P_2))$ — equilibrium constant
4. $k_r = k_f/k_p$ — reverse rate
5. `return` $k_f C[R_1]C[R_2] - k_r C[P_1]C[P_2]$

6.5 duplicateReaction(dbl T, dbl RT, vec C, int recNum, int R1, int R2, int P1, int P2)

Purpose: Calculate the reaction rate for a duplicate bimolecular reaction where k_f is the sum of two Arrhenius rates (reactions `recNum` and `recNum+1`).
Inputs: Same as `reaction`.
Steps:

1. Decrement `recNum` for 0-based array indexing
2. $k_{fa} = A_{rec} \cdot \exp(-E_{a,rec}/RT)$, $k_{fb} = A_{rec+1} \cdot \exp(-E_{a,rec+1}/RT)$; $k_f = k_{fa} + k_{fb}$
3. $k_p = \exp(\texttt{gibbs}(T, R_1, P_1, \&R_2, \&P_2))$; $k_r = k_f/k_p$
4. `return` $k_f C[R_1]C[R_2] - k_r C[P_1]C[P_2]$

6.6 reaction14(dbl T, dbl RT, vec C)

Purpose: Calculate the reaction rate for reaction 14 ($H_2O + H_2O \leftrightarrows H + OH + H_2O$), where H_2O serves as both reactant and third body.
Inputs: Temperature (T), $R_u T$ (RT), species concentrations (C).
Steps:

1. $k_f = A_{14} \cdot T^n \cdot \exp(-E_{a,14}/RT)$
2. $k_p = \exp(\texttt{gibbs}(T, H_2O, H, \texttt{nullptr}, \&OH))$
3. $k_c = 10^{-6} \cdot k_p \cdot P_{\mathrm{ref}}/RT$ — unit correction to mol/cm^3; $k_r = k_f/k_c$
4. `return` $k_f C[H_2O]^2 - k_r C[H]\,C[OH]\,C[H_2O]$

6.7 thirdBodyReaction2R(dbl T, dbl RT, vec C, vec eff, int recNum, int R1, int R2, int P1)

Purpose: Calculate the reaction rate for a third body reaction of the form $R_1 + R_2 + M \leftrightarrows P_1 + M$.
Inputs: Temperature (T), $R_u T$ (RT), species concentrations (C), third body efficiencies (eff), reaction number (recNum), reactant indices (R1, R2), product index (P1).
Steps:

1. $M = \sum_i \alpha_i C_i$ — weighted third body concentration
2. k_f via modified Arrhenius; $k_p = \exp(\texttt{gibbs}(T, R_1, P_1, \&R_2))$
3. $k_c = 10^6 \cdot k_p \cdot RT/P_{\mathrm{ref}}$ — unit correction to cm^3/mol; $k_r = k_f/k_c$
4. `return` $M\big(k_f C[R_1]C[R_2] - k_r C[P_1]\big)$

6.8 thirdBodyReaction2P(dbl T, dbl RT, vec C, vec eff, int recNum, int R1, int P1, int P2)

Purpose: Calculate the reaction rate for a third body reaction of the form $R_1 + M \leftrightarrows P_1 + P_2 + M$.
Inputs: Temperature (T), $R_u T$ (RT), species concentrations (C), third body efficiencies (eff), reaction number (recNum), reactant index (R1), product indices (P1, P2).
Steps:

1. $M = \sum_i \alpha_i C_i$
2. k_f via modified Arrhenius; $k_p = \exp(\texttt{gibbs}(T, R_1, P_1, \texttt{nullptr}, \&P_2))$
3. $k_c = 10^{-6} \cdot k_p \cdot P_{\mathrm{ref}}/RT$ — unit correction to mol/cm^3; $k_r = k_f/k_c$
4. `return` $M\big(k_f C[R_1] - k_r C[P_1]C[P_2]\big)$

6.9 falloffReaction15(dbl T, dbl RT, vec C, vec eff, int R1, int R2, int P1)

Purpose: Calculate the reaction rate for pressure-dependent reaction 15 ($H{+}O_2(+M) \leftrightarrows HO_2(+M)$) using the Troe falloff formulation. Reaction form: $R_1 + R_2 + M \leftrightarrows P_1 + M$.
Inputs: Temperature (T), R_uT (RT), species concentrations (C), third body efficiencies (eff), reactant indices (R1, R2), product index (P1).
Steps:

1. k_0, k_∞ via Arrhenius at low/high pressure limits; $M = \sum_i \alpha_i C_i$
2. $P_r = k_0 M/k_\infty$; $F_c = (1-a)\exp(-T/T_3) + a\exp(-T/T_1)$
3. Compute Troe constants C, N, d; find F via $\log_{10}(F) = \log_{10}(F_c) \,/\, \left[1 + \left((\log_{10} P_r + C)\,/\,(N - d(\log_{10} P_r + C))\right)^2\right]$
4. $k_{eff} = k_\infty P_r F/(1+P_r)$; $k_c = 10^6\, k_p\, RT/P_{\text{ref}}$; $k_r = k_{eff}/k_c$
5. `return` $k_{eff}C[R_1]C[R_2] - k_r C[P_1]$

6.10 falloffReaction22(dbl T, dbl RT, vec C, vec eff, int R1, int P1, int P2)

Purpose: Calculate the reaction rate for pressure-dependent reaction 22 ($H_2O_2(+M) \leftrightarrows OH + OH(+M)$) using the Troe falloff formulation. Reaction form: $R_1 + M \leftrightarrows P_1 + P_2 + M$.
Inputs: Temperature (T), R_uT (RT), species concentrations (C), third body efficiencies (eff), reactant index (R1), product indices (P1, P2).
Steps:

1. k_0, k_∞, M, P_r, F_c, F computed identically to `falloffReaction15` using reaction-22 Arrhenius constants
2. $k_{eff} = k_\infty P_r F/(1+P_r)$; $k_c = 10^{-6}\, k_p\, P_{\text{ref}}/RT$; $k_r = k_{eff}/k_c$
3. `return` $k_{eff}C[R_1] - k_r C[P_1]C[P_2]$

6.11 globalRates(dbl T, vec C)

Purpose: Compute the reaction rate for every active reaction in the Burke mechanism and return them as a vector.
Inputs: Temperature (T), species concentrations (C).
Steps:

1. $RT = R_uT$
2. Call `reaction`, `duplicateReaction`, `thirdBodyReaction2R/2P`, `reaction14`, `falloffReaction15/22` for each of the 21 active reactions
3. `return` q of length 27, with omitted reactions (3, 7, 8, 10, 11, 21, 27) set to `0.0`

6.12 heatRelease(vec& q, dbl T)

Purpose: Convert reaction rates to volumetric heat release (W/m^3) by multiplying each rate by its net reaction enthalpy.
Inputs: Reaction rates (q) from `globalRates`, Temperature (T).
Steps:

1. For each active reaction i: $q_i \mathrel{*}= \texttt{enthalpy}(T, \ldots) \leftarrow q_i$ becomes $\dot{q}_i$ in W/cm^3
2. $\dot{q} = \sum_i \dot{q}_i \times 10^6$ — unit conversion from cm^3 to m^3
3. `return` $\dot{q}$ (W/m^3)

6.13 massSource(vec q)

Purpose: Calculate species mass source terms (kg/m^3·s) from the reaction rates.
Inputs: Reaction rates (q) from `globalRates`.
Steps:

1. Compute molar production rate $\dot{s}_k = \sum_i \nu_{i,k}\, q_i$ for each species using stoichiometric coefficients ν
2. $S_k = M_{w,k} \cdot \dot{s}_k \times 10^3$ — convert mol/cm^3·s to kg/m^3·s
3. `return` S (skipping N_2 at index 2)

7 Implementation

Source terms are computed in `computeModelSources`, which is called once per advection timestep. Rather than computing instantaneous rates and applying them directly, chemistry is integrated cell-by-cell over the full advection interval Δt_{adv} using a constant-volume explicit Euler ODE sub-stepping loop. This decouples the stiff chemistry timescale from the larger fluid timestep imposed by ICE.

7.1 Step 1: Build the Cell State / Bulletproofing

For each cell retrieve from the data warehouse: Δt_{adv}, temperature T, density ρ, and the 8 tracked species mass fractions. Compute the N_2 closure fraction and insert it at index 2 to form the full 9-species vector:

$$\mathbf{Y} = [Y_{H_2},\ Y_{O_2},\ Y_{N_2},\ Y_{H_2O},\ Y_H,\ Y_O,\ Y_{OH},\ Y_{HO_2},\ Y_{H_2O_2}], \qquad Y_{N_2} = 1 - \sum_{k \neq N_2} Y_k$$

Bulletproofing checks are applied: $\rho > 0$, $100 < T < 5000$ K (hard limit), $300 < T < 3500$ K (NASA-7 polynomial validity warning), and $Y_k \geq 0$ for all species.

7.2 Step 2: Constant-Volume ODE Integration ($t = 0 \to \Delta t_{\text{adv}}$)

Starting from Δt_{chem} s, the following sub-steps repeat until $t = \Delta t_{\text{adv}}$. The final sub-step is clamped so the loop ends exactly at Δt_{adv}.

7.2.1 2a: Mixture Specific Heat

Non-dimensional species specific heats $\hat{c}_{p,k}$ are evaluated from the NASA 7 polynomials via **cp-SpecificHeat**. Mixture averages are assembled as:

$$c_{p,\text{mix}} = \sum_k Y_k R_k \hat{c}_{p,k}, \qquad R_k = \frac{R_u}{M_{w,k}} \quad \text{[J/kg-K]}$$
$$R_{\text{mix}} = \sum_k Y_k R_k$$

Constant-volume specific heat and ratio of specific heats follow from ideal gas relations:

$$c_v = c_{p,\text{mix}} - R_{\text{mix}}, \qquad \gamma = \frac{c_{p,\text{mix}}}{c_v}$$

7.2.2 2b: Molar Concentrations

Mass fractions are converted to molar concentrations (mol/cm^3):

$$C_k = 10^{-3}\,\frac{\rho\, Y_k}{M_{w,k}}$$

7.2.3 2c: Reaction Rates (f = globalRates)

globalRates(T, $\mathbf{C}$) returns the rate vector $\boldsymbol{\omega}$ (mol/cm^3·s) for all 21 active reactions.

7.2.4 2d: Species Integration (f = massSource)

massSource($\boldsymbol{\omega}$) returns the mass source vector $\mathbf{S}$ (kg/m^3·s). Each tracked species is advanced one sub-step:

$$Y_k^{n+1} = Y_k^n + \Delta t_{\text{chem}} \frac{S_k}{\rho}$$

N_2 is updated from closure after each sub-step. The cumulative species source term is accumulated as:

$$\text{massSrc}_k \mathrel{+}= \frac{S_k\,\Delta t_{\text{chem}}}{\rho}$$

7.2.5 2e: Energy Integration (f = heatRelease)

heatRelease($\boldsymbol{\omega}$, T) returns the volumetric heat release rate $\dot{q}$ (W/m^3). Temperature is advanced one sub-step under constant-volume assumption:

$$T^{n+1} = T^n + \Delta t_{\text{chem}} \frac{\dot{q}}{\rho\, c_v}$$

The cumulative energy source term (J) is accumulated as:

$$\text{engSrc} \mathrel{+}= \dot{q}\cdot V_{\text{cell}}\cdot \Delta t_{\text{chem}}$$

7.3 Step 3: Write Source Terms

After the integration loop completes, the accumulated terms are added to the ICE source arrays:

- $Y_{\text{src},k} \mathrel{+}= \text{massSrc}_k$ (dimensionless mass fraction source)
- `eng_src` += engSrc (J)

7.4 Thermodynamic Properties for ICE (modifyThermoTransportProperties)

Separately from the source term calculation, `modifyThermoTransportProperties` is scheduled to run before each advection step. It loops over all cells (including ghost cells) and evaluates $c_{p,\mathrm{mix}}$, R_{mix}, c_v, and γ from the current temperature and species mass fractions using the same NASA polynomial procedure as Step 2a. The results are written directly to ICE's `specific_heatLabel` (c_v) and `gammaLabel` (γ). ICE reads these labels during equation of state evaluation, so the mixture thermodynamic properties remain consistent with the evolving composition throughout the simulation.